Code No: 136CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
MOBILE COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Compare and contrast the system architectures of GSM and GPRS. [2]
b) Demonstrate the process of handover in mobile communications. [3]
c) List the types of Medium Access Control (MAC) techniques used in wireless networks. [2]
d) Demonstrate the differences between FDMA and TDMA in the context of wireless communication. [3]
e) Demonstrate understanding of database adaptation by outlining client-server computing techniques. [2]
f) Explain the process of query processing in the context of database issues. [3]
g) Define Data Synchronization and provide an example protocol. [2]
h) Outline the key features of Classification of Data Delivery Mechanisms. [3]
i) Summarize the applications of Mobile Agents in MANETs. [2]
j) Outline the features of protocols WAP and Bluetooth in mobile computing. [3]

PART – B

(50 Marks)

2.a) How would you plan and develop a novel mobile application that addresses the limitations of current handheld devices?
b) Analyze the motives and causes behind the challenges faced in mobile computing paradigms, providing evidence to support your generalizations. [5+5]

OR

3.a) Evaluate and judge the importance of security protocols in GSM systems, defending your opinion based on a set of criteria.
b) Propose an alternative architecture for mobile communications, combining elements in a new pattern to maximize efficiency and overcome current impediments. [5+5]

4.a) Develop a plan to address near and far terminal challenges in a wireless environment, experimenting with novel techniques to improve communication reliability.
b) Dissect the components of Location Management in Mobile Networks, identifying relationships between registration and tunneling processes. [5+5]

OR

5.a) Assess the importance of DHCP in wireless networks, determining its influence on IP address assignment.

b) Propose an original model for handling dynamic changes in network traffic by inventing a solution that maximizes spectrum utilization in a Wireless LAN using TDMA. [5+5]

6.a) Analyze the motives behind utilizing Mobile TCP in mobile networks and contrast its features with Conventional TCP/IP Protocols.

b) Evaluate the importance of Database Hoarding and Caching Techniques in client-server computing, defending their impact on transactional models and QoS issues. [5+5]

OR

7.a) Apply knowledge of Indirect TCP in mobile transport layer protocols to propose alternative solutions for optimizing data recovery processes in mobile networks.

b) Create a plan to adapt and modify existing transport layer protocols for mobile networks, considering the challenges posed by Snooping TCP and other protocols.[5+5]

8.a) Develop a plan to maximize the efficiency of data delivery mechanisms in a given scenario.

b) Assess the impact of software and protocols on the effectiveness of data synchronization. [5+5]

OR

9.a) Justify the selection of a specific data dissemination method based on a set of criteria.

b) Design an original approach to overcome challenges in communication asymmetry during data dissemination. [5+5]

10.a) Categorize the functionalities of protocols such as WAP, J2ME, and Linux for Mobile Devices, and analyze how they contribute to the overall success of mobile computing.

b) Justify and defend the selection of specific routing algorithms for different MANET applications based on a set of criteria related to performance, reliability, and adaptability. [5+5]

OR

11.a) Develop a comprehensive plan for integrating multiple mobile computing platforms to create a seamless and adaptable environment for diverse mobile applications.

b) Experiment with Mobile Agents and propose innovative ways to utilize them for efficient data transfer and processing within a MANET environment. [5+5]

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